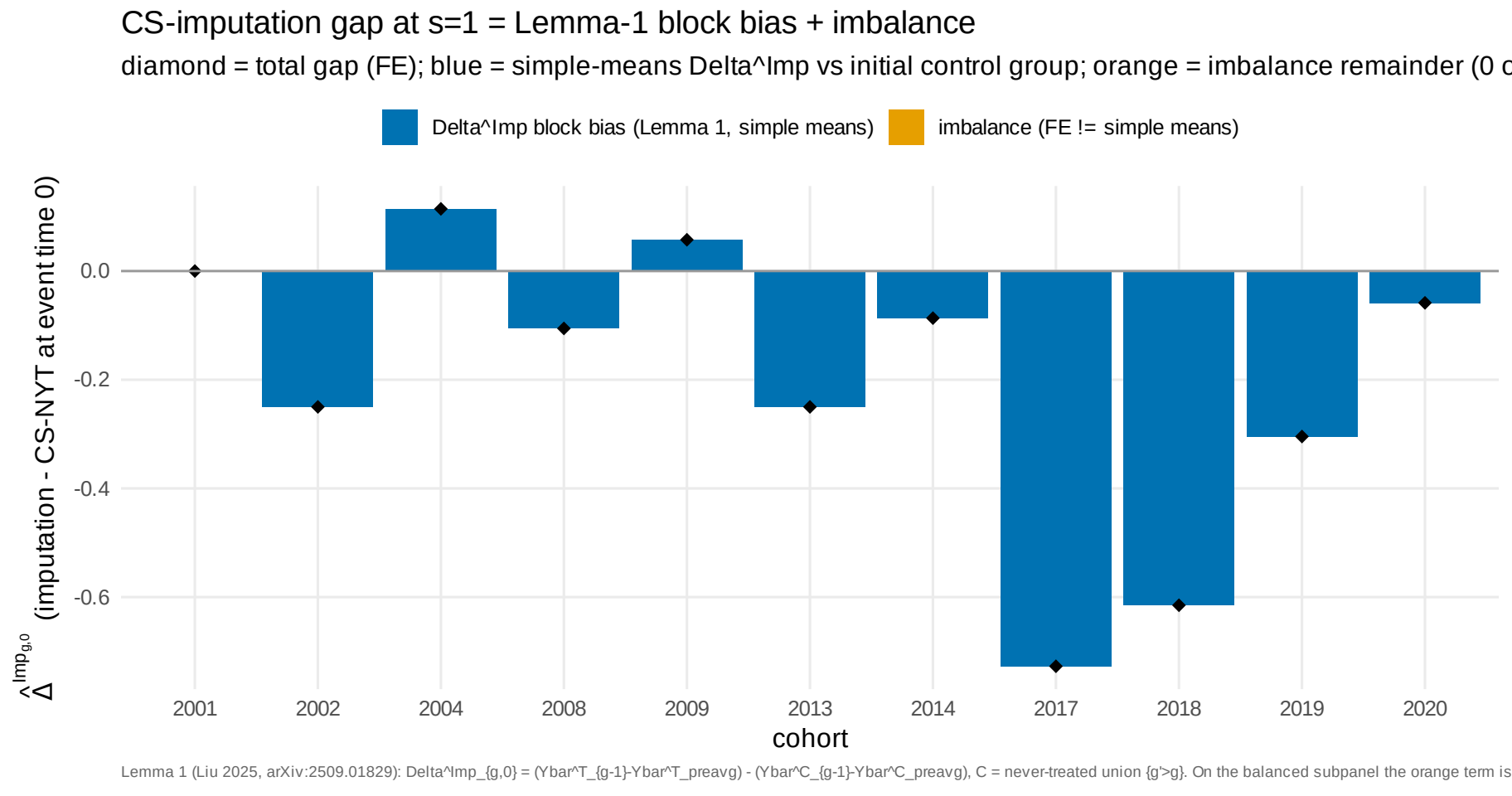
CS-imputation gap at s=1 = Lemma-1 block bias + imbalance
diamond = total gap (FE); blue = simple-means Delta^Imp vs initial control group; orange = imbalance remainder (0 c
Delta^Imp block bias (Lemma 1, simple means)
imbalance (FE != simple means)
(imputation - CS-NYT at event time 0)
0.0
-0.2
-0.4
-0.6
2001
2002
2004
2008
2009
2013
2014
2017
2018
2019
2020
cohort
Lemma 1 (Liu 2025, arXiv:2509.01829): Delta^Imp_{g,0} = (Ybar^T_{g-1}-Ybar^T_preavg) - (Ybar^C_{g-1}-Ybar^C_preavg), C = never-treated union {g'>g}. On the balanced subpanel the orange term is